

PROJECT SENTINEL

User Documentation

Microsoft 365 Governance & Security Intelligence

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6.1 Glossary

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Appendix A: Output Field Reference

A1. audit_logs_signin

A2. file_metadata_summary

A3. sharing_permissions

1. Introduction

1.1 Scope:

Project Sentinel is a Microsoft 365 governance monitoring solution developed as part of the La Trobe University Industry Development Program (IDP) 2026, in partnership with SECMON1. The project was initiated in response to a recognised industry need as organisations adopt Microsoft 365 at scale, the complexity of managing permissions, access controls, data governance, and compliance grows significantly. Manual auditing of these areas is time consuming, error prone, and difficult to maintain consistently.

Sentinel addresses this by automating the collection and analysis of governance data directly from the Microsoft 365 tenant. It runs structured extraction modules against the tenant's APIs, produces standardised JSON output files, and surfaces key governance findings through a SharePoint hosted dashboard all without requiring any external infrastructure or local software installations.

1.2 What does Sentinel monitor?

Sentinel runs five active data extraction modules, each targeting a specific governance domain within Microsoft 365. Each module produces a full JSON file (complete record-level detail) and a SUMMARY JSON file (aggregated totals and governance flags).

Module Name	Governance Domain	What It Collects
audit_logs_signin	Identity & Access	Sign-in events - user identity, application used, IP address, device details (OS, browser, compliance), location(city, state, country), success/failure status, risk state, and whether Conditional Access was applied.
file_metadata_summary	Data Governance	Files in SharePoint and OneDrive - file name, type (.docx/.xlsx/.pdf), size, sensitivity label applied (or absence of label), and sharing state.
sharing_permissions	Sharing& Collaboration	All permission records - who has access to what, at what level, whether any links are anonymous, and whether any external (guest) users are present.
purview_labels	Classification& Protection	Sensitivitylabel assignments per file from Microsoft Purview-label name, label level, and whether protection is applied.
dlp_extractor	DataLoss Prevention	DLP policy match records - policy name, detection rule, match count, and violation indicators per file.

Table 1.2: Sentinel Extraction Modules and Governance Domains

1.3 Compliance Framework

The following Microsoft Purview compliance policies are set up in the sentinels.onmicrosoft.com tenant alongside Sentinel:

Policy / Feature	Type	Current Configuration
Sensitivity Labels	Information Protection	4 labels: Public (0), Internal (1), Confidential (2), Highly Confidential (3). Scope: Files & other data assets.
Sentinel-DLP-Policy	Data Loss Prevention	Rule: Sentinel-Sensitive Data-Rule. Covers Exchange email, SharePoint sites, OneDrive. Mode: Simulation (1 match found). Email notifications: On.
Sentinel-1Year-Policy	Retention	Retention period: 1 year (delete after). Locations: Exchange, SharePoint, OneDrive, Microsoft 365 Group mailboxes & sites. Status: Enabled.
Unified Audit Log	Audit & Monitoring	Active - searches performed via Microsoft Purview → Audit.

Table 1.3: Microsoft Purview Compliance Policies Configured in Tenant

2. Getting Started

2.1 Prerequisites

Requirement	Details
Microsoft 365 Account	A valid account in the sentinals.onmicrosoft.com tenant.
SharePoint Group Membership	Must be added to one of the three Sentinel Workspace permission groups. Without this, you will see an Access Denied error.
Supported Web Browser	Microsoft Edge (recommended) or Google Chrome.
Internet Connection	Stable outbound HTTPS access to Microsoft 365.

Table 2.1: Access Requirements and Prerequisites

2.2 Accessing the sentinel-workspace SharePoint Site

1. Open Microsoft Edge or Google Chrome.
2. Navigate to the Sentinel-Workspace SharePoint site URL provided by your administrator.
3. Sign in with your Microsoft 365 account (username@sentinals.onmicrosoft.com).
4. Complete any multi-factor authentication (MFA) prompt if applicable.
5. You will be taken to the Sentinel-Workspace home page.

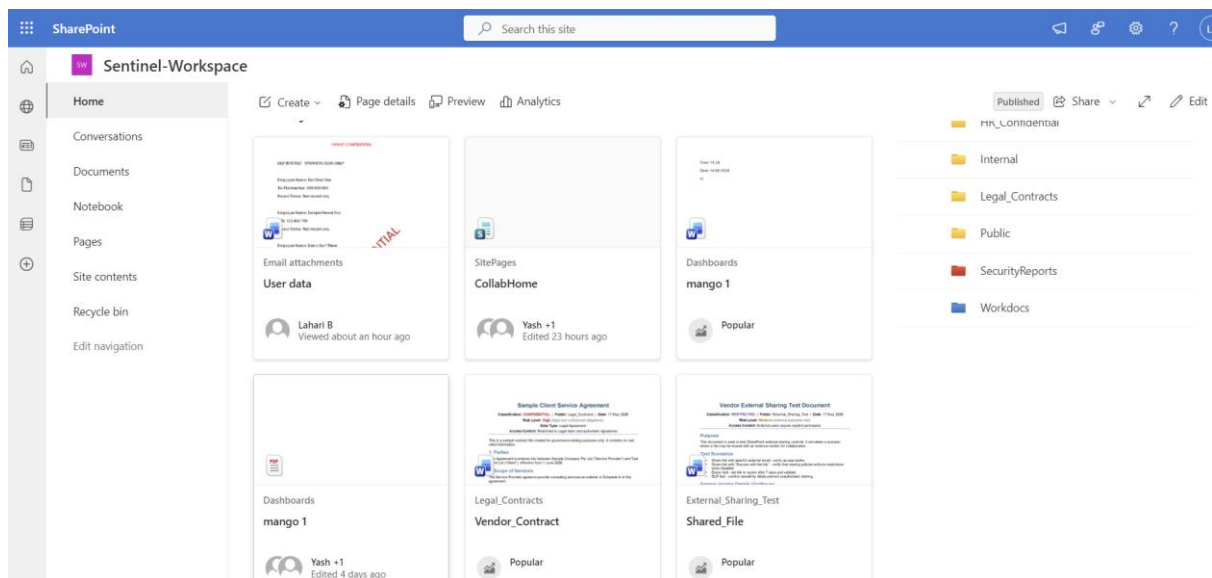


Fig: 2.2 - Sentinel-Workspace SharePoint Site Home Page

2.3 SharePoint Permission Group

Group Name	Permission Level	Who It Is For
Sentinel-Workspace Owners	Full Control	Administrators - manage site settings, add/remove users, upload and delete files, and publish pages.
Sentinel-Workspace Members	Edit	Active team members - upload output files, edit page content, and update lists.
Sentinel-Workspace Visitors	Read	Stakeholders and reviewers - view findings and documents only. Cannot make changes.

Table 2.3 SharePoint Permission Groups and Access Levels

2.4 Navigating the Site

Navigation Item	What You Will Find There
Home	The site landing page - summary information and links to key resources.
Documents	Document library containing all Sentinel JSON output files, named with module name and timestamp.
Pages	Contains the Sentinel governance dashboard page - the primary interface for viewing findings visually.
Notebook	Shared OneNote notebook for team notes and observations.
Site contents	Full list of all libraries, lists, and apps. Used by administrators for site management.

Table 2.4: Sentinel-Workspace Site Navigation Items

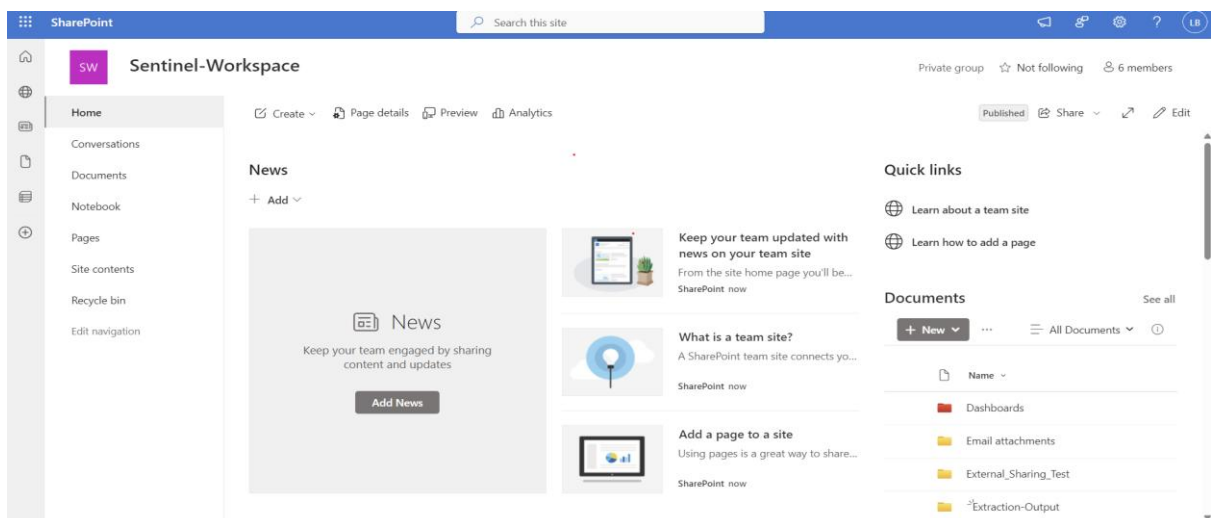


Fig: 2.4 Sentinel-Workspace Site Navigation Panel

2.5 Understanding the Document Library

The Documents library is the main storage area in the Sentinel-Workspace site. To open it, click Documents in the left navigation panel. The library has two types of content: the governance test data folders and the Extracted-Output folder where the scan output files are saved.

1. Dashboards: contains files related to the governance dashboard built by the team.
2. Email_attachments: stores files shared through the group mailbox linked to the Sentinel Workspace site.
3. External_Sharing_Test: this folder was created during testing to check how external sharing behaves within the tenant. It appears in the sharing permissions scan results.
4. Extracted_Output: this is the main folder for Sentinel scan results. After every scan, the nine output files are saved here automatically by the extraction script.

Governance Test Data Folders

The Documents library contains several folders that were set up as a test environment for governance analysis. Sample files were uploaded to represent realistic organisational data across different sensitivity levels. These are the folders that the Sentinel scan modules run against:

- HR_Confidential - employee personal records and HR-related documents. High sensitivity.
- Finance_Confidential - salary and financial files. High sensitivity.
- Legal_Contracts - legal documents and contracts.
- Internal - general team files for internal use only.
- Public - files intended to be shared openly.
- External_Sharing_Test - files used to test external sharing behaviour.
- SecurityReports - governance analysis reports produced by the security analyst.
- Workdocs - general working documents used by the team.

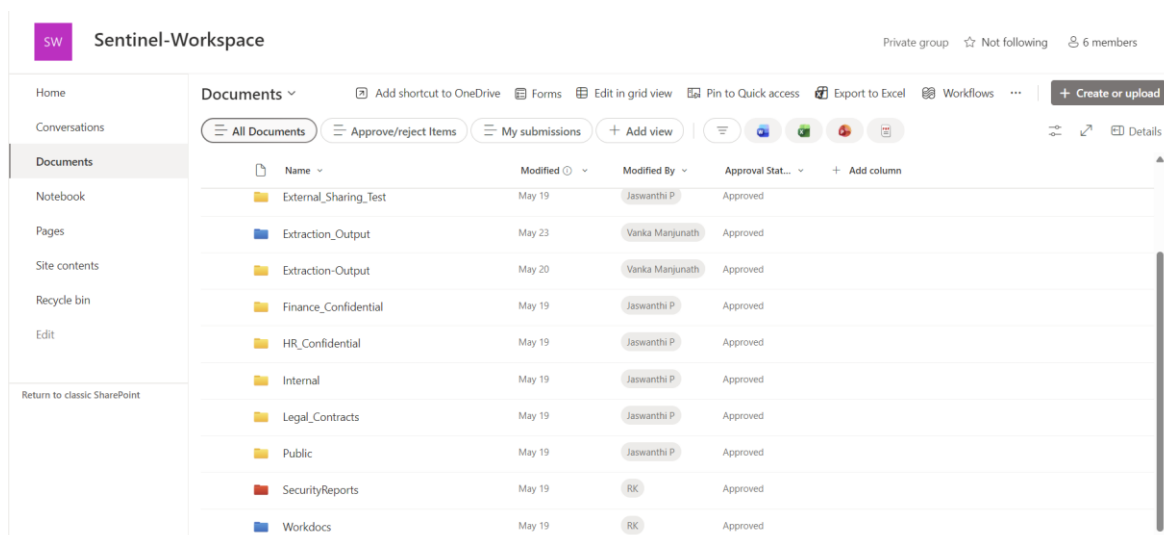


Fig: 2.5.1 Governance Test Data Folders

Extracted-Output Folder

The Extracted-Output folder is where Vanka Manjunath uploads the JSON output files after each scan run. Inside it there are five subfolders, one for each type of scan output:

- audit - sign-in audit log files from the audit_logs_signin module.
- dlp - Data Loss Prevention scan result files.
- files - file metadata output from the file_metadata_summary module.
- permissions - sharing permissions output from the sharing_permissions module.
- timeline - timeline records from the scan run.

Each scan produces nine output files in total: two files per extraction module (a full record level JSON and a summary JSON with totals and governance flags), plus a consolidated extraction history file. The files are named using the module name plus the date and time the scan ran, so it is easy to match files from the same scan:

Format: <module_name>_<YYYYMMDD>_<HHMMSS>.json

<module_name>_SUMMARY_<YYYYMMDD>_<HHMMSS>.json

Examples: audit_logs_signin_20260518_073754.json

audit_logs_signin_SUMMARY_20260518_073754.json

file_metadata_20260518_073818.json

file_metadata_summary_SUMMARY_20260518_073818.json

sharing_permissions_20260518_073829.json

sharing_permissions_SUMMARY_20260518_073829.json

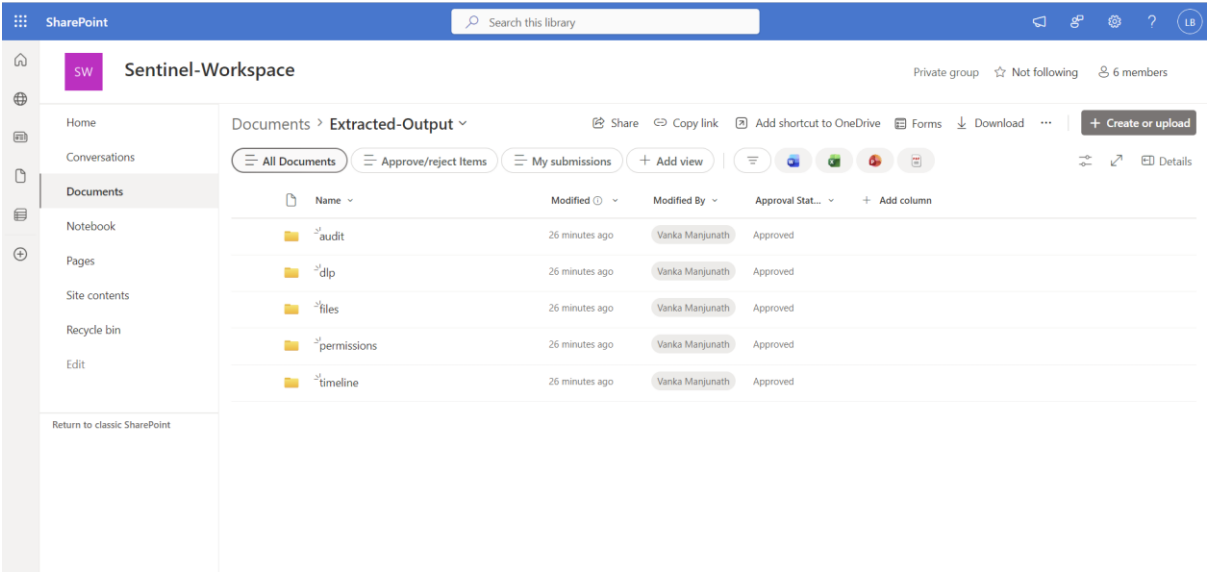


Fig: 2.5.2 Documents Library and Extracted-Output Folder Structure

3. Using the tool

3.1 How sentinel Runs a Scan

Sentinel scans are run by an administrator. The scripts log in to the Microsoft 365 tenant, run each module one at a time, and save the output JSON files to the Documents library on SharePoint. Regular users do not need to start a scan themselves. They just open the SharePoint site or the dashboard after a scan has been run to see the latest results.

What happens during a scan

1. The administrator starts the extraction script.
2. The script authenticates to the Microsoft 365 tenant using MSAL (Microsoft Authentication Library) with a certificate via the Azure App Registration.
3. The script enumerates SharePoint sites, document drives, and users.
4. The audit_logs_signin module runs - Graph API collects all sign-in events from the unified audit log.
5. The file_metadata_summary module runs - Graph API collects file names, types, sizes, and sensitivity labels from SharePoint and OneDrive.
6. The sharing_permissions module runs - Graph API collects all permission records across SharePoint drives.
7. Six JSON output files are saved to the Extracted-Output folder in the Documents library, each with a timestamp in the file name.

3.2 Understanding the output files

Each scan run produces multiple JSON and CSV output files across five extraction modules plus a consolidated extraction history file. Files are named using the module name and the UTC timestamp of the scan run. The key files for governance review are the SUMMARY files -these contain totals and governance flags without needing to open the full record files.

File	Contents & Key Fields
audit_logs_signin_[ts].json	Full sign-in records: id, createdDateTime, userDisplayName, userPrincipalName, appDisplayName, ipAddress, clientAppUsed, conditionalAccessStatus, riskState, status.errorCode, deviceDetail (OS, browser, isCompliant), location (city, state, country).

audit_logs_signin_SUMMARY_[ts].json	Totals: total_records, successful_logins, failed_logins, risky_signins. Flags: high_failure_rate, risky_signin_present.
audit_logs_directory_[ts].json	Directory activity audit records - user and group changes, role assignments, and admin operations.
audit_logs_file_activity_[ts].json	File-level activity records - who opened, edited, shared, or deleted files and when.
file_metadata_[ts].json	Full file records: name, type (.docx/.xlsx/.pdf), size in bytes, sensitivity label (or Unlabelled), sharing state, site URL, last modified date.
file_metadata_summary_SUMMARY_[ts].json	Totals: total_files, total_size_bytes, file_type_breakdown, sensitivity_label_breakdown, shared_files_count, shared_files_pct.
sharing_permissions_[ts].json	Full permission records: resource, grantee, access level, anonymous link flag, external user flag.
sharing_permissions_SUMMARY_[ts].json	Totals: total_permissions, anonymous_links, external_users. Flags: anonymous_links_present, high_external_exposure.
purview_labels_SUMMARY_[ts].json	Purview label scan results: total_labels_configured, label names detected.
governance_timeline_[ts].json	Consolidated timeline combining file, permission, and sign-in events in chronological order for audit traceability.
extraction_history.xlsx	Running history of all scan runs - module name, record count, output file name, status, and timestamp for each extraction.

Table 3.2: Output Files Produced per Extraction Run

File naming

[ts] in the file names above stands for the UTC timestamp of the scan run in the format YYYYMMDD_HHMMSS.

Example: audit_logs_signin_20260521_032337.json was produced on 21 May 2026 at 03:23:37 UTC.

All files from the same scan run share the same timestamp, so you can match them together easily.

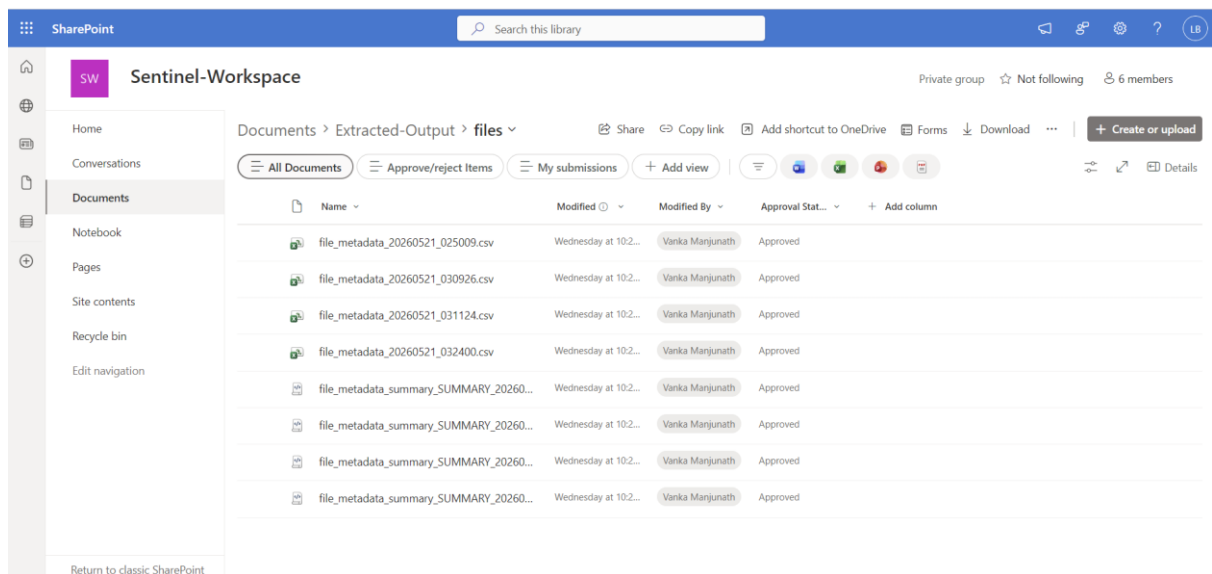


Fig: 3.2 Extracted-Output Folder with Timestamped JSON Files

3.3 Latest scan details - 21 May 2026

The table below shows the findings from the most recent Sentinel scan run (21 May 2026, 03:24 UTC). Numbers are taken directly from the SUMMARY JSON files produced during the scan.

Module	Finding	Value / Status
audit_logs_signin	Total records	354
audit_logs_signin	Successful logins	302 (85.3%)
audit_logs_signin	Failed logins	52 (14.7%)
audit_logs_signin	Risky sign-ins	0 (Governance flag: CLEAR)
file_metadata_summary	Total files	26 (.docx: 11 / .xlsx: 9 / .pdf: 6)
file_metadata_summary	Sensitivity labels	26 out of 26 files labelled (all files: Internal)
file_metadata_summary	Shared files	26 out of 26 (100%)
sharing_permissions	Total permissions	105
sharing_permissions	Anonymous links	0 (Governance flag: CLEAR)
sharing_permissions	External users	0 (Governance flag: CLEAR)

Table 3.3: Latest Scan Results - 21 May 2026

3.4 Using the Dashboard

The dashboard was developed using Microsoft Power BI Desktop, as shown in the screenshots. The visuals and data models were created locally and can be published to the Power BI Service for sharing and scheduled refresh.

While SharePoint is mentioned as the hosting platform, the dashboard itself is a Power BI report that can be embedded into SharePoint Online for organizational access. The scheduled refresh (e.g., every Monday after 6 AM) applies when the report is deployed to the Power BI Service environment.

- **Audit Monitoring** - sign-in volume, most active users, login activity over time, and file sharing exposure distribution.
- **File Insights** - total files, shared files overview, password protection status, and file link type breakdown.
- **Governance and Compliance** - total shared records and governance exposure breakdown across the tenant.
- **Tenant Overview** - four key metrics at a glance: total files, total users, high-risk files, and total audit events.
- **File Activity Dashboard** - file activity trend over time, user activity distribution, file access distribution, and a user slicer for drilling into a specific account.

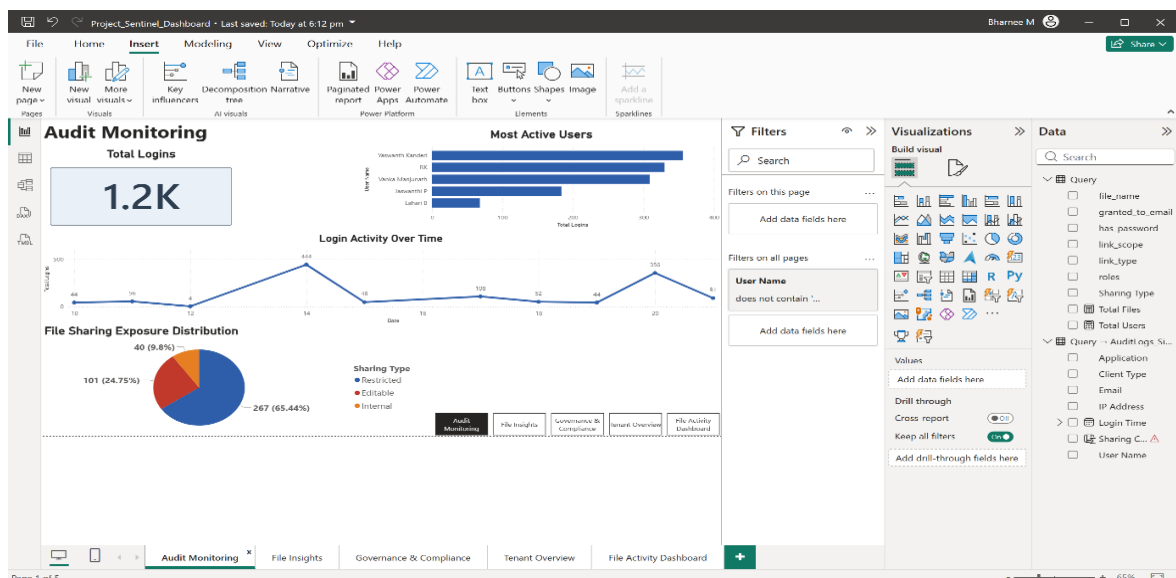


Fig: 3.4.1 Dashboard Tab 1: Audit Monitoring

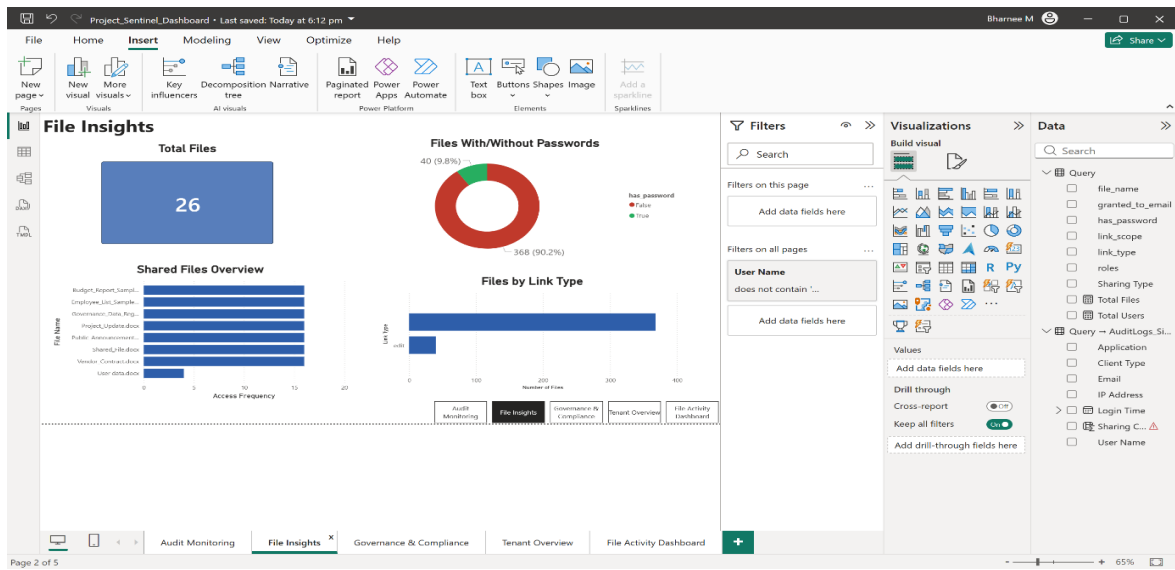


Fig: 3.4.2 Dashboard Tab 2: File Insights

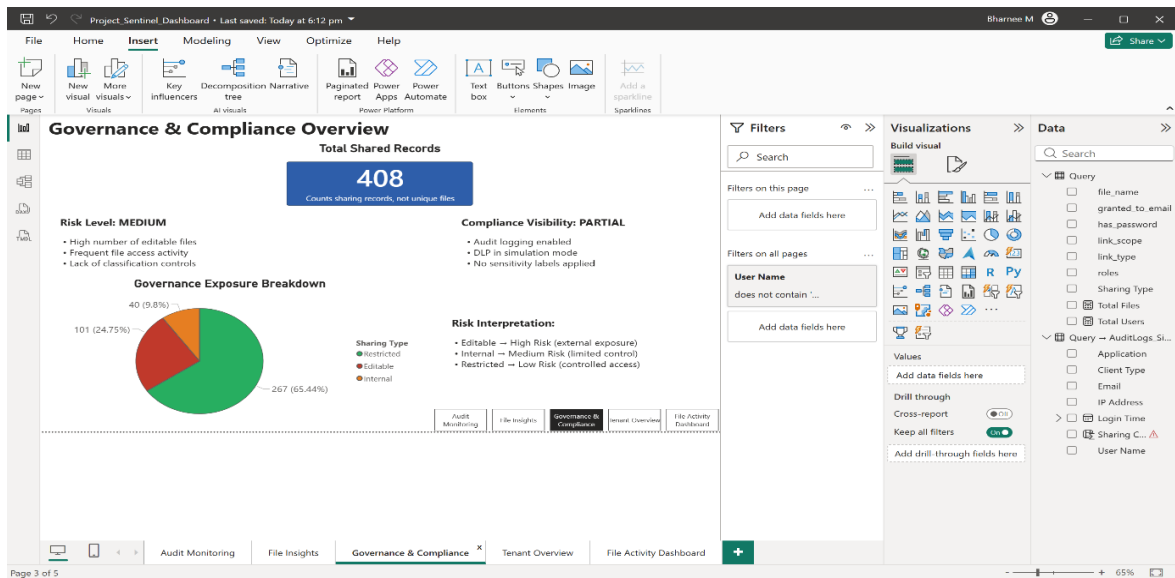


Fig: 3.4.3 Dashboard Tab 3: Governance and Compliance

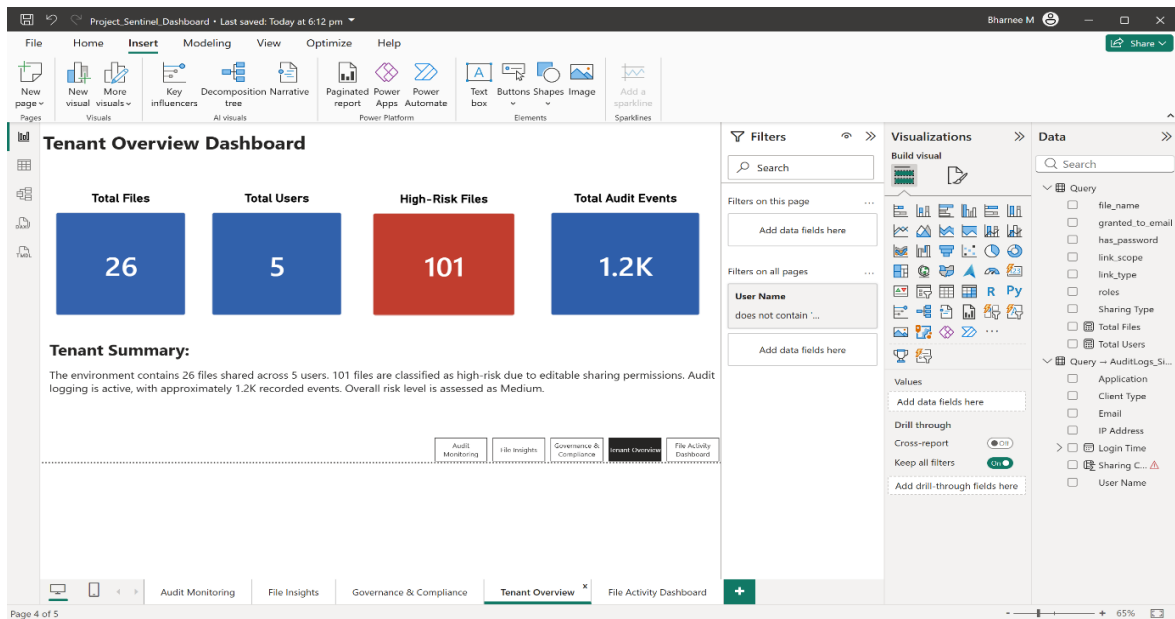


Fig: 3.4.4 Dashboard Tab 4: Tenant Overview

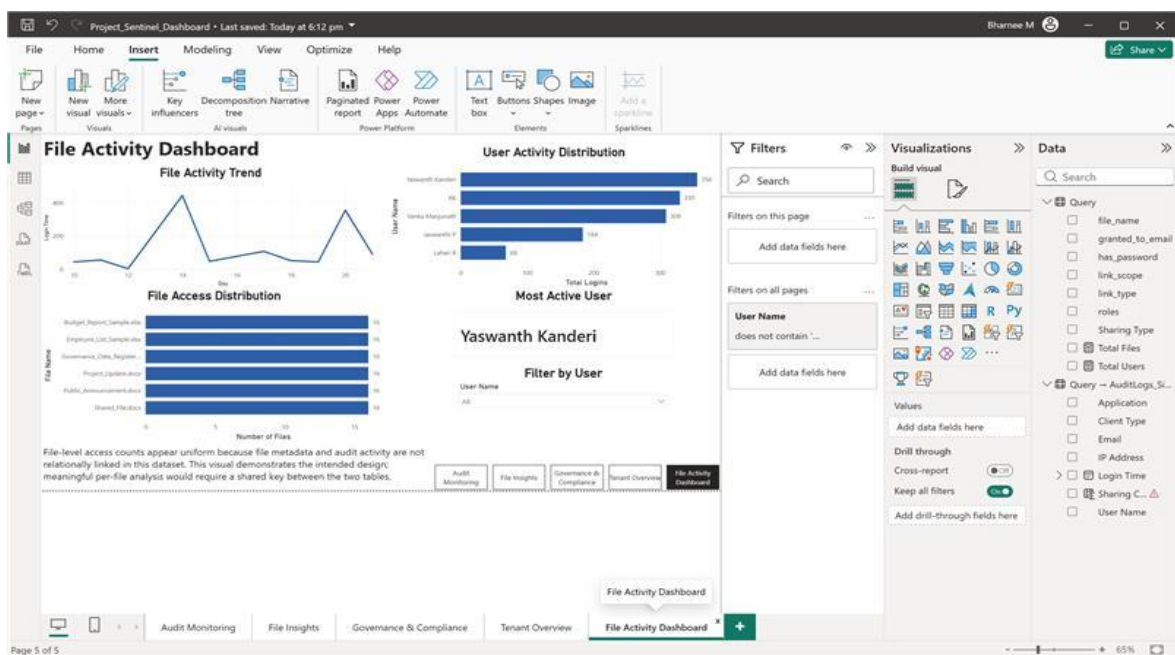


Fig: 3.4.5 Dashboard Tab 5: File Activity Dashboard

3.5 Governance Risk Findings

Based on the latest scan, the following governance risks were identified across the five extraction modules. Each risk is rated by severity and includes the evidence from the scan data and a recommended action.

Risk Area		Description	Evidence	Risk	Recommendation
Sensitivity Classification		All files carry the Internal sensitivity label, providing a baseline classification. Higher-sensitivity folders (HR_Confidential, Finance_Confidential) should be reviewed for Confidential or Highly Confidential labelling.	26 of 26 files labelled as Internal.	HIGH	Apply labels to all files using the four labels configured in Purview, starting with business-critical content.
Excessive File Sharing		All files are shared, meaning unclassified content is accessible beyond a single user, which increases the data exposure risk.	26 of 26 files shared while all carry the Internal sensitivity label.	MEDIUM	Review sharing scope for each file and restrict access to required users. Combine with sensitivity labels once applied.
Weak DLP Enforcement		The DLP policy runs in simulation mode only, so violations are detected and logged but not actively prevented.	DLP policy configured; mode = Simulation; 1 match detected during testing.	MEDIUM	Move the DLP policy from simulation to enforcement mode after confirming the simulation match is not a false alarm.
Broad Internal Access		Group-based permissions give Members edit access by default, which may grant modification rights more widely than needed.	105 total permissions; Members hold Edit access, Visitors hold Read access.	MEDIUM	Review group membership and move users to Visitor where edit rights are not required. Schedule periodic access reviews.
Authentication Risk (Failed Logins)		A moderate number of failed sign-in attempts is recorded, which may reflect credential errors or access probing.	52 failed logins out of 354 audit records (14.7%).	LOW	Monitor failed-login trends, enforce MFA, and investigate any spikes or repeated failures

				against specific accounts.
Lack of Threat Detection	Audit logging provides retrospective visibility but there is no advanced threat detection or real-time alerting in place.	354 audit records captured; 0 risky sign-ins flagged; no advanced detection configured.	MEDIUM	Enable risk-based detection such as Entra ID Protection to move from reactive monitoring to proactive threat oversight.

Table 3.3 Governance Risk Findings and Recommended Actions

3.6 Microsoft Purview - Sensitivity Labels

Label	Priority	When to Apply
Public	0 - Lowest	Publicly shareable content - marketing materials, published reports.
Internal	1	Internal team use only - project documents, meeting notes, internal reports.
Confidential	2	Sensitive business information - financial data, client information, strategic plans.
Highly Confidential	3 - Highest	Strictly restricted - legal documents, security credentials, executive communications.

Table 3.4: Sensitivity Label Guide

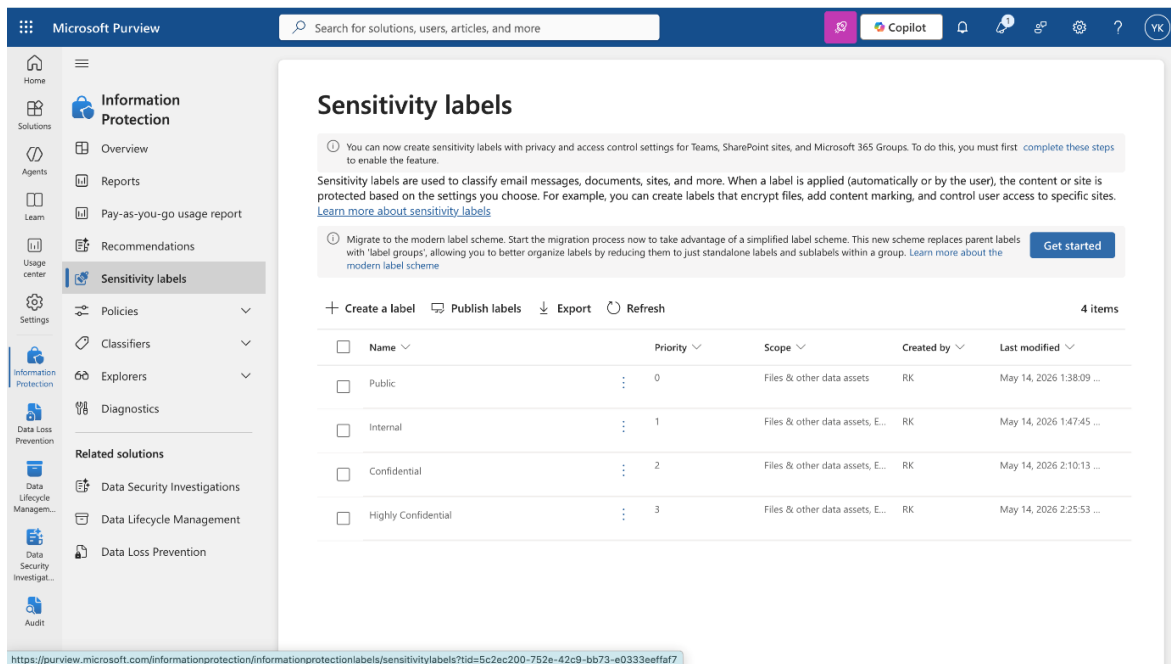


Fig: 3.6 Microsoft Purview Sensitivity Labels Configuration

3.7 Microsoft Purview- DLP and Retention Policies

Sentinel - DLP - Policy

This policy is configured to check for sensitive data across Exchange, SharePoint sites, and OneDrive accounts. It runs the Sentinel-SensitiveData-Rule to detect sensitive content types. The policy is currently in simulation mode, which means it flags and logs matches but does not block anything. One match was detected during simulation testing, which confirmed the detection rule is working correctly. The team should review the simulation results before switching the policy to active enforcement.

Feature	Status	Notes
Policy name	Sentinel-DLP-Policy	DLP policy for Project Sentinel SharePoint and OneDrive test data.
Detection rule	Sentinel-SensitiveData-Rule	Configured to detect sensitive information types across the tenant.
Locations covered	Exchange email - All accounts, SharePoint sites, OneDrive accounts - All accounts	Covers all email and file storage channels in the tenant.
Email notifications	On	Administrators are notified by email when a policy match is detected.

Policy mode	In simulation (Review matching items)	Monitors and reports violations. Does not block or prevent anything yet.
Simulation result	1 match found	One match was detected during simulation testing, confirming the rule is functioning correctly.

Table 3.5: Sentinel-DLP-Policy Configuration Details

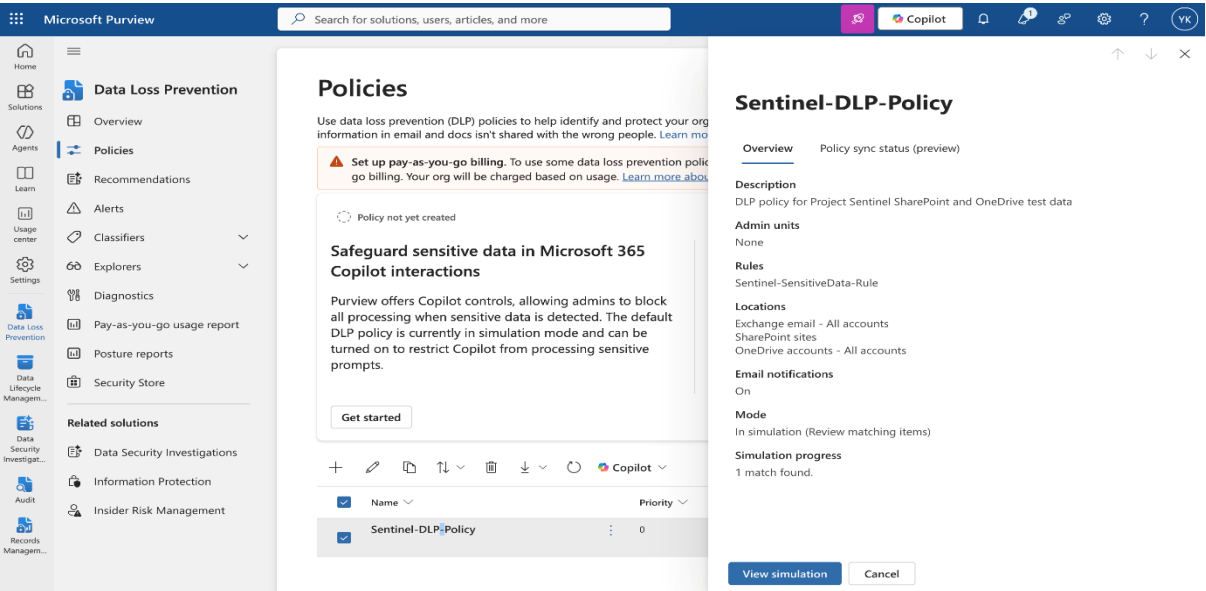


Fig: 3.7.1 Microsoft Purview Sensitivity Labels Configuration

Sentinel - 1Year - Policy

This policy keeps all content across Exchange mailboxes, SharePoint classic and communication sites, OneDrive accounts, and Microsoft 365 Group mailboxes and sites for one year, then deletes it. The status shows as Enabled (Pending), which means it is still spreading across all the locations in the tenant. This is expected behaviour and resolves on its own within 24 to 48 hours. No action is needed. Preservation lock is set to No.

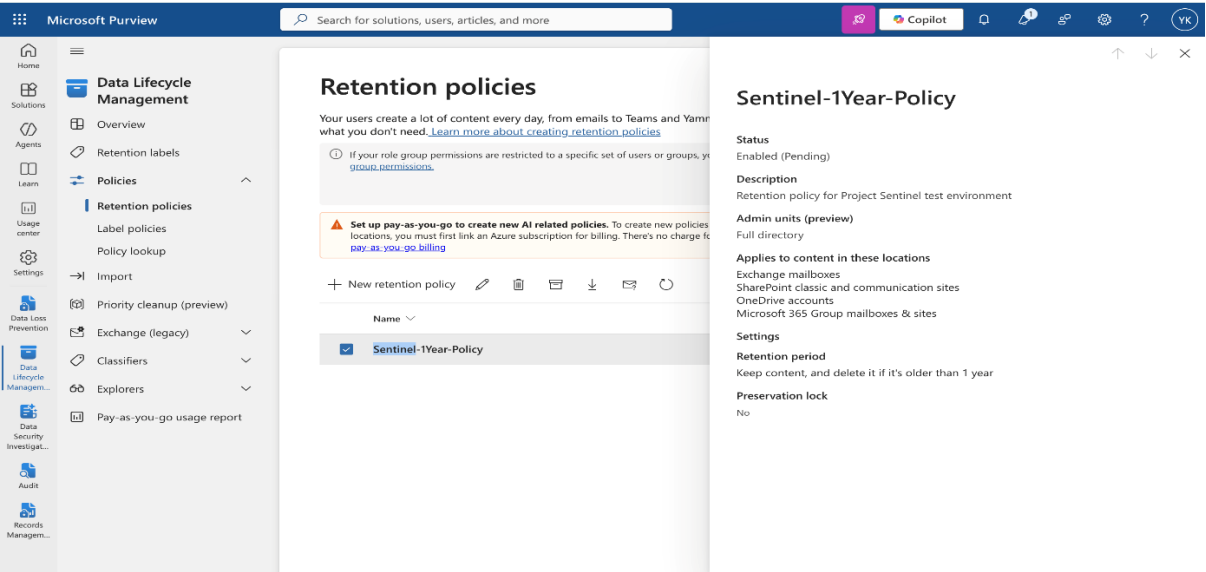


Fig: 3.7.2 Sentinel-1Year-Policy Retention Configuration

3.8 Step - by - Step: Applying a Sensitivity Label

1. Navigate to the SharePoint document library.
2. Hover over the file name and click the checkbox to select it.
3. Click the three-dot menu (...) → Details.
4. In the Details panel, scroll to Sensitivity label and click the dropdown.
5. Select the appropriate label: Public, Internal, Confidential, or Highly Confidential.
6. Click Save.

3.9 Common User Scenarios

The following scenarios walk through the most common tasks that different users carry out in Sentinel. Basic scenarios cover everyday actions any team member might perform. Advanced scenarios cover tasks that require deeper access or judgement, typically carried out by the security analyst or administrator.

Basic Scenarios

Scenario 1: Checking the latest governance findings

Role: Compliance Officer or Stakeholder.

Goal: Review the latest scan findings to check the current governance status of the tenant.

1. Open the Sentinel-Workspace SharePoint site and click on Documents in the left navigation.
2. Open the Extracted-Output folder and navigate to the audit, files, or permissions subfolder depending on what you want to review.
3. Open the most recent SUMMARY JSON file - it is the one with the latest timestamp in the file name.
4. Check the governance_flags section. If any flag shows true, note it and refer to Section 3.5 for the recommended action.

Scenario 2: Applying a sensitivity label to a file

Role: Any team member with edit access.

Goal: Apply the correct sensitivity label to a file in the Documents library so it is classified for the next scan.

1. Open the Sentinel-Workspace Documents library and locate the file you want to label.
2. Hover over the file name, click the three-dot menu, and select Details.

3. In the Details panel, find Sensitivity label, click the dropdown, and choose the correct label based on the file content. Refer to Section 3.6 for guidance on which label to pick.
4. Click Save. The label will be picked up by the file_metadata_summary module in the next scan.

Scenario 3: Checking who has access to the site

Role: Administrator.

Goal: Verify which users are in each permission group to ensure only authorised people have access.

1. Go to the Sentinel-Workspace site and click the Settings gear in the top right.
2. Click Site permissions. You will see the three groups: Sentinel-Workspace Owners, Members, and Visitors.
3. Click each group to see the members. Remove anyone who no longer needs access or move them to a lower permission group.

Advanced Scenarios

Scenario 1: Investigating a spike in failed logins

Role: Security Analyst.

Goal: A scan shows a high number of failed logins. Investigate which accounts are affected and whether it is a genuine threat.

1. Open the latest audit_logs_signin_SUMMARY JSON file from the Extracted-Output/audit folder. Check the failed_logins count and whether high_failure_rate is set to true.
2. Open the full audit_logs_signin JSON file from the same folder. Look for records where status.errorCode is non-zero. Note the userPrincipalName, ipAddress, and location for each failure.
3. Go to Microsoft Purview and run an Audit Log Search (Section 4.5) for the affected account over the same date range. This gives a full picture of all activity for that user.
4. If failures are concentrated on a single account from an unusual location or IP, escalate to the Team Lead (RK) to consider blocking or resetting that account in the Microsoft 365 Admin Centre.

Scenario 2: Running an audit log search for a specific user

Role: Security Analyst or Administrator.

Goal: Pull all Microsoft 365 activity for a specific user over a given time period to investigate unusual behaviour.

1. Sign in to Microsoft Purview (<https://purview.microsoft.com>) and go to Audit in the left panel.
2. Click Search, set the date range for the period you want to investigate, and enter the user's UPN in the Users field (e.g., username@sentinals.onmicrosoft.com).
3. Name the search, click Search, and wait for the job to complete. Click the search name to view results. Export if needed for a formal report.

Scenario 3: Moving the DLP policy from simulation to enforcement

Role: Administrator. Goal: The simulation results have been reviewed and confirmed. Move Sentinel-DLP-Policy to active enforcement so violations are blocked rather than just logged.

- Sign in to Microsoft Purview and go to Data Loss Prevention, then Policies. Click Sentinel-DLP-Policy.
- Before making the change, review the simulation matches. Click View simulation results and confirm no legitimate content was flagged incorrectly.
- Click Edit policy, go to the Policy mode page, select Turn the policy on immediately, review the summary, and click Submit.
- After activation, run a new Sentinel scan and check the DLP section in the next SUMMARY file to confirm the policy is now active.

4. Administrator Guide

4.1 Deployment Overview

Sentinel uses a Python-based extraction engine that connects to the Microsoft 365 tenant through the Microsoft Graph API. Authentication is handled via an Azure App Registration using a certificate - no passwords or client secrets are stored in the code. The extraction runs four steps: authenticate, enumerate resources, extract metadata, and save structured JSON output to SharePoint.

Component	Role
Azure App Registration	Connects Sentinel to the Microsoft 365 tenant using certificate-based authentication. Application permissions are used so the extraction runs without any user interaction.
Python Extraction Engine (MSAL)	Core scripts. Uses Microsoft Authentication Library (MSAL) to authenticate, then runs the three extraction modules one at a time.
Microsoft Graph API	Primary data collection interface. Permissions configured: Sites.Read.All, AuditLog.Read.All, Directory.Read.All, InformationProtectionPolicy.Read, Files.Read.All.
Sentinel-Workspace SharePoint Site	Central hub - stores JSON output files in the Extracted-Output folder, hosts the dashboard, and controls access via three permission groups.
Microsoft Purview	Compliance layer - sensitivity labels, Sentinel-DLP-Policy (simulation mode, 1 match), Sentinel-1Year-Policy (retention), and audit log searches.

Table 4.1: System Components and Their Roles

4.2 Running a Scan

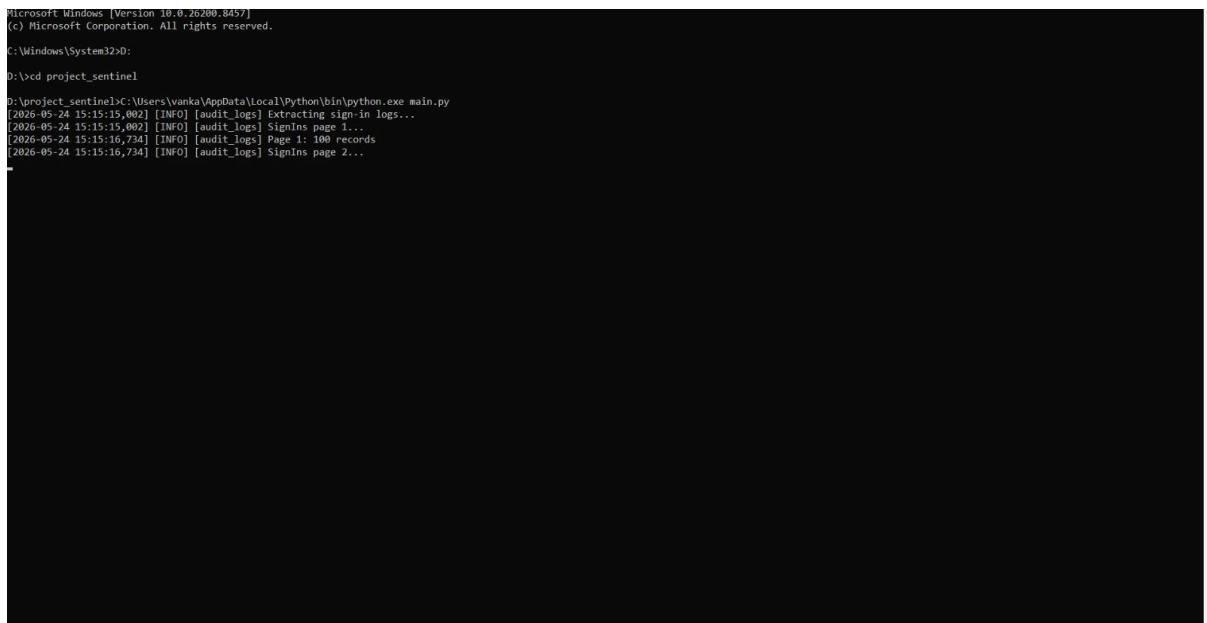
Scans are run by using the Python extraction scripts. The script uses certificate-based authentication and does not require any passwords to be entered each time. Below are the steps that happen when a scan is run:

1. The API Developer opens a terminal, navigates to the backend folder, and runs:
python main.py
2. The script authenticates to the Microsoft 365 tenant using MSAL and the Azure App Registration certificate.
3. The script enumerates all SharePoint sites, document drives, and users in the tenant.

4. The five extraction modules run one at a time - audit_logs_signin, file_metadata_summary, sharing_permissions, purview_labels, and dlp_extractor.
5. Each module makes paginated Graph API calls to collect all records. Large datasets are handled automatically using the @odata.nextLink pagination mechanism.
6. The output is normalised into structured JSON format with UTC timestamps.
7. Six JSON files (two per module - one full, one summary) are uploaded to the Extracted-Output folder in the SharePoint Documents library.

After Running a Scan

1. Go to the Documents library and check that six new JSON files have appeared in the Extracted-Output subfolders with today's date and time.
2. Open the three SUMMARY files and check for any governance flags that are set to true.
3. If there are any HIGH risk flags, look into them within 48 hours.
4. Open the dashboard and check that the findings have updated to match the new scan.



```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>D:

D:\>cd project_sentinel

D:\project_sentinel>C:\Users\vanka\AppData\Local\Python\bin\python.exe main.py
[2026-05-24 15:15:15,002] [INFO] [audit_logs] Extracting sign-in logs...
[2026-05-24 15:15:15,002] [INFO] [audit_logs] Signins page 1...
[2026-05-24 15:15:16,734] [INFO] [audit_logs] Page 1: 100 records
[2026-05-24 15:15:16,734] [INFO] [audit_logs] Signins page 2...
```

Fig: 4.2.1 Extraction Script Running in Terminal

```

Administrator: Command Prompt
[2026-05-24 15:17:05,379] [INFO] [file_metadata] [Files[Documents]] Total records: 4
[2026-05-24 15:18:06,166] [INFO] [file_metadata] [Drives[My workspace]] Fetching page 1
[2026-05-24 15:18:09,080] [WARNING] [file_metadata] Skipping drives for site My workspace: HTTP 403 Forbidden
[2026-05-24 15:18:09,082] [INFO] [file_metadata] [Drives[Designer]] Fetching page 1
[2026-05-24 15:18:10,820] [WARNING] [file_metadata] Skipping drives for site Designer: HTTP 403 Forbidden
[2026-05-24 15:18:10,824] [INFO] [file_metadata] [Drives[Team Site]] Fetching page 1
[2026-05-24 15:18:10,893] [INFO] [file_metadata] [Drives[Team Site]] Total records: 1
[2026-05-24 15:18:10,895] [INFO] [file_metadata] [Files[Documents]] Fetching page 1
[2026-05-24 15:18:11,454] [INFO] [file_metadata] [Files[Documents]] Total records: 0
[2026-05-24 15:18:11,457] [INFO] [file_metadata] [Drives[Communication site]] Fetching page 1
[2026-05-24 15:18:11,940] [INFO] [file_metadata] [Drives[Communication site]] Total records: 1
[2026-05-24 15:18:11,942] [INFO] [file_metadata] [Files[Documents]] Fetching page 1
[2026-05-24 15:18:12,353] [INFO] [file_metadata] [Files[Documents]] Total records: 0
[2026-05-24 15:18:12,360] [INFO] [file_metadata] [Drives[Group for Answers in Viva Engage - DO NOT DELETE 185275482112]] Fetching page 1
[2026-05-24 15:18:13,340] [INFO] [file_metadata] [Drives[Group for Answers in Viva Engage - DO NOT DELETE 185275482112]] Total records: 1
[2026-05-24 15:18:13,343] [INFO] [file_metadata] [Files[Documents]] Fetching page 1
[2026-05-24 15:18:13,741] [INFO] [file_metadata] [Files[Documents]] Total records: 0
[2026-05-24 15:18:13,743] [INFO] [file_metadata] [Drives[La Trobe]] Fetching page 1
[2026-05-24 15:18:14,740] [INFO] [file_metadata] [Drives[La Trobe]] Total records: 1
[2026-05-24 15:18:14,743] [INFO] [file_metadata] [Files[Documents]] Fetching page 1
[2026-05-24 15:18:15,796] [INFO] [file_metadata] [Files[Documents]] Total records: 1
[2026-05-24 15:18:15,797] [INFO] [file_metadata] [Files[Documents]] Fetching page 1
[2026-05-24 15:18:16,833] [INFO] [file_metadata] [Files[Documents]] Total records: 0
[2026-05-24 15:18:16,835] [INFO] [file_metadata] [Drives[All Company]] Fetching page 1
[2026-05-24 15:18:17,488] [INFO] [file_metadata] [Drives[All Company]] Total records: 1
[2026-05-24 15:18:17,489] [INFO] [file_metadata] [Files[Documents]] Fetching page 1
[2026-05-24 15:18:18,471] [INFO] [file_metadata] [Files[Documents]] Total records: 0
[2026-05-24 15:18:18,481] [INFO] [file_metadata] File Metadata Extraction Complete
[2026-05-24 15:19:52,722] [INFO] [purview_labels] Starting Purview Label Extraction
[2026-05-24 15:19:52,722] [INFO] [purview_labels] Extracting sensitivity labels from Purview...
[2026-05-24 15:19:53,116] [WARNING] [purview_labels] Could not extract sensitivity labels: HTTP 400
[2026-05-24 15:19:53,118] [INFO] [purview_labels] Extracting label policies from Purview...
[2026-05-24 15:19:53,693] [WARNING] [purview_labels] Could not extract label policies: HTTP 400
[2026-05-24 15:19:53,696] [INFO] [purview_labels] Purview Label Extraction Complete
[2026-05-24 15:19:53,697] [INFO] [dlp_extractor] Starting DLP Extraction
[2026-05-24 15:19:53,697] [INFO] [dlp_extractor] Extracting DLP policies...
[2026-05-24 15:19:54,049] [WARNING] [dlp_extractor] Could not extract DLP policy: HTTP 400
[2026-05-24 15:19:54,052] [INFO] [dlp_extractor] Extracting sensitivity labels...
[2026-05-24 15:19:54,945] [WARNING] [dlp_extractor] Could not extract sensitivity labels: HTTP 400
[2026-05-24 15:19:54,947] [INFO] [dlp_extractor] Extracting information protection config...
[2026-05-24 15:19:55,789] [INFO] [dlp_extractor] Protection config retrieved
[2026-05-24 15:19:55,792] [INFO] [dlp_extractor] DLP Extraction Complete
[2026-05-24 15:19:56,696] [INFO] [timeline] Building consolidated timeline...
[2026-05-24 15:19:56,697] [INFO] [timeline] Timeline built - 873 events
Synced 366 files to OneDrive successfully
Extraction complete. Excel history updated: output\extraction_history.xlsx
D:\project_sentinel>

```

Fig: 4.2.2 Scan Output Files Saved to SharePoint After Run

4.3 Managing the Sentinel - Workspace Site

Adding Users to Permission Groups

1. Sign in as an Owner → Settings gear → Site permissions.
2. Click the appropriate group (Owners / Members / Visitors).
3. Click Invite people → enter email → Add.

Managing the Documents Library

- After each scan, check that nine output files have been saved with the correct date and time.
- Move files older than 30 days into a dated subfolder (for example, /2026-05/) to keep the library organised.
- The library uses the same permissions as the parent site. Do not change this unless a specific file needs to be restricted to certain people.

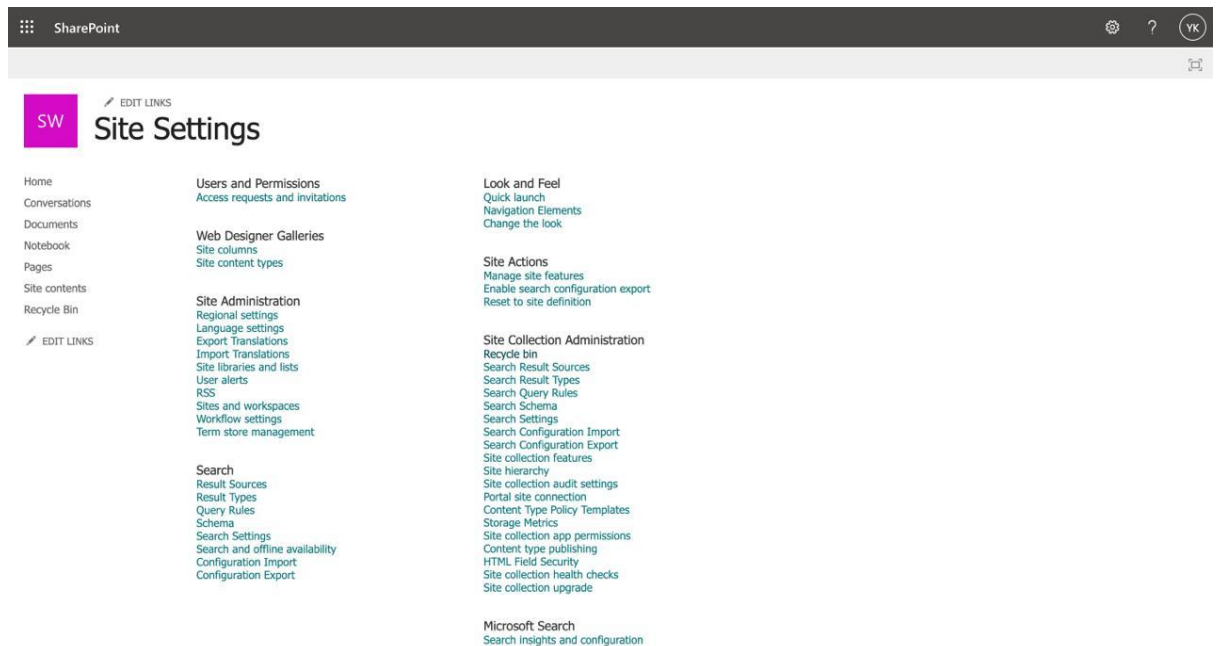


Fig: 4.3 SharePoint Site Permission Groups Management

4.4 Managing Microsoft Purview Compliance Policies

Moving DLP Policy to Enforcement Mode

1. Microsoft Purview → Data Loss Prevention → Policies → Sentinel-DLP-Policy → Edit policy.
2. On the Policy mode page, change from simulation to Turn the policy on immediately.
3. Review the summary and click Submit.

Retention Policy Monitoring

The Sentinel-1Year-Policy will show as Enabled (Pending) while it is still spreading across all the locations in the tenant. This can take 24 to 48 hours. Once it switches to Enabled (Active), nothing else needs to be done.

4.5 Running Audit Log Searches

The Purview Audit Search tool lets you search through all Microsoft 365 activity logs in the tenant. As of May 2026, the team has run 28 audit searches, with results covering daily periods. The most recent search (May 20 to May 21) returned 1031 audit events. Searches are performed by Yaswanth (YaswanthKanderi@Sentinals.onmicrosoft.com).

1. Sign in to Microsoft Purview (<https://purview.microsoft.com>) and click Audit in the left panel.
2. Click Search in the left menu.

3. Set the Date and time range (UTC) - enter the Start date and End date you want to search.
4. Optionally filter by Activities, Users, Workloads, or enter a keyword to narrow results.
5. Give the search a name in the Search name field so you can find it later.
6. Click Search and wait for the Job status to show Completed.
7. Click the search name to view or export the results

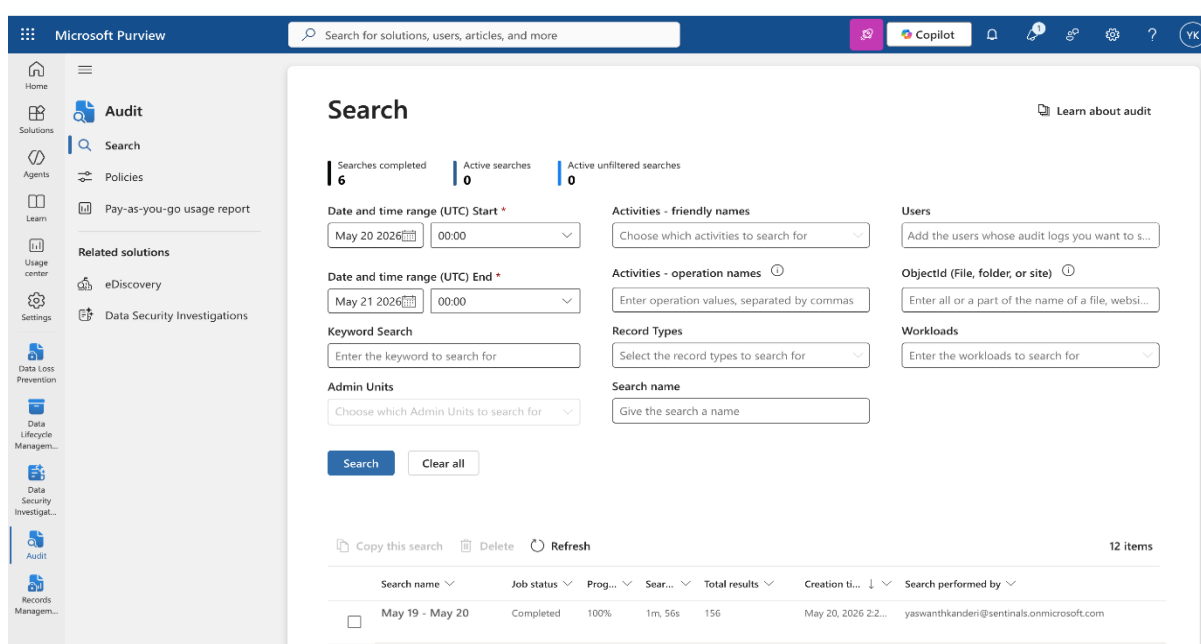


Fig: 4.5 Microsoft Purview Audit Log Search Interface

4.6 Administrator Security Responsibilities

- Check the permission groups every 30 days and remove anyone who no longer needs access.
- After every scan, check the SUMMARY files for any governance flags. If a flag is true, it should be investigated within 48 hours.
- Make sure all files in the Documents library have a sensitivity label applied.
- Look at the DLP simulation results carefully before turning the policy to enforcement mode.
- Keep all scan output JSON files for at least 12 months. They are the main audit record for the governance work done.

4.7 Administrator Maintenance Checklist

Frequency	Task	Description
After each scan	Review SUMMARY files	Check all three SUMMARY JSON files for governance flag changes. Escalate HIGH flags.
After each scan	Confirm dashboard refresh	Open dashboard and verify it reflects the new scan timestamp.
Weekly	Run audit search	Run a Microsoft Purview audit search covering the past 7 days.
Monthly	Review permission groups	Remove users who no longer need access.
Monthly	Check DLP simulation	Review new matches. Prepare for enforcement activation.
Monthly	Archive old output files	Move files older than 30 days to a dated archive subfolder.
Quarterly	Sensitivity label audit	Check file_metadata SUMMARY for unlabelled files.

Table 4.7: Administrator Maintenance Checklist

5. Known Issues & Tips

Issue	Likely Cause	Resolution
Output files missing after scan	Script failed or lacks write permission.	Check with Owners that scan completed. Confirm the script account has Edit permission on Documents library.
Dashboard shows old data	Browser cache.	Hard refresh: Ctrl+Shift+R (Windows) or Cmd+Shift+R (Mac).
Access Denied on site	User not in any permission group.	Contact an Owner to be added to Visitors or Members group.
Files labelled as Internal only	All files carry Internal. HR_Confidential and Finance_Confidential may warrant higher labels.	Review high-sensitivity folders and update labels to Confidential or Highly Confidential where appropriate per Section 3.8.
Audit search no results	Date range too narrow or filters too restrictive.	Broaden the date range and clear all optional filters. Logs can take up to 24 hours to appear.
Retention policy shows Enabled (Pending)	Policy still propagating - expected behaviour.	No action required. Resolves automatically within 24–48 hours.
DLP simulation shows unexpected matches	Rule matching legitimate content.	Review in Purview → Sentinel-DLP-Policy → View simulation. Refine rule before activating enforcement.

Table 5: Known Issues and Resolutions

5.1 General Tips

Tip:

Always start with the SUMMARY files. They give a quick picture of the governance status without needing to go through all the records.

Keep a record of when each scan was run and what the governance flags showed. This helps to spot any changes or patterns over time.

Before labelling files in bulk, check with the team first and agree on which label applies to which type of content.

If a sign-in looks unusual, use Microsoft Purview Audit Search to look at the full activity for that account.

6. Glossary and Version History

6.1 Glossary

Term	Definition
Admin Consent	Approval by a Global Administrator allowing an application to access tenant data on behalf of the organisation.
Audit Log	Chronological record of Microsoft 365 actions. Sentinel extracts sign-in audit logs via audit_logs_signin module.
Conditional Access	Azure AD policies enforcing MFA or device compliance based on sign-in conditions.
DLP (Data Loss Prevention)	Purview feature detecting sensitive information. Sentinel-DLP-Policy monitors Exchange, SharePoint, and OneDrive.
Documents Library	SharePoint library in Sentinel-Workspace where all JSON output files are stored after each scan.
Governance Flag	Boolean in a SUMMARY JSON indicating a risk threshold was exceeded (e.g., anonymous_links_present: true).
Graph API	Microsoft's unified REST API for Microsoft 365. Sentinel's primary data collection interface.
Guest User	External account granted limited access to the tenant. Monitored via sharing_permissions module.
JSON	Structured output format used by Sentinel. Each module produces a full JSON and a SUMMARY JSON per scan.
Microsoft Purview	Microsoft compliance platform - sensitivity labels, DLP, retention, and audit log search.
MFA(Multi-FactorAuthentication)	Sign-in security requiring two or more verification factors.
Retention Policy	Purview policy specifying content retention duration. Sentinel-1Year-Policy retains all content for one year.
Risky Sign-In	Sign-in flagged by Identity Protection as potentially compromised.

Sensitivity Label	Classification tag for files and emails. Four labels: Public, Internal, Confidential, Highly Confidential.
Sentinel-Workspace	SharePoint site serving as Sentinel's central hub - output files, dashboard, and permission groups.
SharePoint REST API	RESTful API for SharePoint Online. Used to collect file metadata and sharing permissions.
Simulation Mode	DLP mode where content is flagged but not blocked. Used to test policy before enforcement.
SUMMARY JSON	Aggregated output file per module per scan. Contains totals and governance flags. Named with _SUMMARY_.
Tenant	Dedicated Microsoft 365 environment. Project Sentinel operates in sentinals.onmicrosoft.com.
UPN (User Principal Name)	User's sign-in ID in Microsoft 365 - e.g., Lahari@sentinals.onmicrosoft.com.

Table 6.1: Glossary of Terms

6.2 Version History

Version	Date	Changes
v1.0	May 2026	Initial release. Three modules: audit_logs_signin, file_metadata_summary, sharing_permissions. SharePoint deployment. Purview integration: 4 sensitivity labels, DLP policy, retention policy.

Table 6.2: Version History

Appendix A: Output Field Reference

Key fields in each module's full JSON output file

A1. audit_logs_signin

Field	Type	Description
id	String (GUID)	Unique identifier for the sign-in event.
createdDateTime	DateTime (UTC)	Timestamp of the sign-in event.
userDisplayName	String	Display name of the user who signed in.
userPrincipalName	String	UPN of the signing user (e.g., Lahari@sentinals.onmicrosoft.com).
appDisplayName	String	Application used for the sign-in.
conditionalAccessStatus	Enum	success / failure / notApplied.
riskState	Enum	none / atRisk / confirmedCompromised / remediated.
status.errorCode	Integer	0 = success. Non-zero = failure with specific reason.
deviceDetail.isCompliant	Boolean	Whether device is enrolled in and compliant with device management policies.
location.countryOrRegion	String	Country code of sign-in origin (e.g., AU).

Appendix A1: Output Field Reference: audit_logs_signinOutput

A.2 file_metadata_summary

Field	Type	Description
total_files	Integer	Total files in the scan scope.
file_type_breakdown	Object	Count per file extension (e.g., {'.docx': 11, '.xlsx': 9, '.pdf': 6}).
sensitivity_label_breakdown	Object	Count per sensitivity label. 'Unlabelled' = no label applied.
shared_files_pct	Float	Percentage of total files that are shared (100.0 = all files shared).

Appendix A2: Output Field Reference: file_metadata_summary

A.3 sharing_permissions

Field	Type	Description
total_permissions	Integer	Total permission records found.
anonymous_links	Integer	Links accessible without sign-in. Should be 0 in a secure environment.
external_users	Integer	External guest users with access to internal resources.
governance_flags.anonymous_links_present	Boolean	True if any anonymous links exist. Triggers immediate review.

Appendix A3: Output Field Reference: sharing_permissions